TEXORA · LECTURE SHEET

Relationship of Fiber, Yarn and Fabric Strength

Strength is lost at every step from fibre to yarn, then partly regained when yarns help each other inside a fabric. This sheet follows that translation chain, explains the bundle effect and fabric assistance, and shows how weave, density and finishing move the final fabric value.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 4 · LEVEL 3 TERM 2

1. The strength translation chain

Every fabric strength figure has a genealogy. The fibre sets a ceiling, spinning converts fibre strength into yarn strength at a substantial loss, and weaving assembles yarns into a structure that — surprisingly — can carry more than the sum of its threads tested alone. The chain is:

Fibre tenacity → yarn tenacity → strength of the threads in a strip → measured fabric strength

Understanding the chain matters because each link has different remedies. A weak fabric caused by a weak fibre needs a different raw material. A weak fabric caused by poor translation in spinning needs process correction. A weak fabric caused by structure — too few threads, an open weave, a damaging finish — needs a construction change. Testing only the fabric tells you that there is a problem; testing along the chain tells you where it lives. All values in this sheet refer to specimens conditioned at 65% RH and 20 °C.

2. From fibre to yarn — why only about half survives

In TE-3103 the reference values for the cotton system used in this course were a fibre tenacity of about 26 g/tex and a resulting yarn tenacity of 12.45 cN/tex. In consistent units the yarn realises roughly half of the fibre tenacity — a translation efficiency of the order of 50%. That loss is not carelessness; it is geometry and statistics:

- **Helix angle.** Twist, which holds the yarn together, lays the fibres in helices. A helical fibre contributes only a component of its strength along the yarn axis, and the component falls as the surface twist angle rises. The yarn needs twist for cohesion, and pays for it in axial strength.

- **Unequal loading.** Fibres in a yarn are not all taut at the same extension. Slack fibres carry nothing until the yarn has extended; over-taut fibres break early and shed their load suddenly. At any moment only part of the cross-section is working at full capacity.
- **Short fibres and fibre ends.** A staple yarn is full of fibre ends. Near an end a fibre is held only by friction over a short length and slips rather than breaks, contributing little. Short-fibre content therefore taxes yarn strength twice: fewer effective fibres, and more weak places.
- **Migration and unevenness.** Fibres migrate between the core and the surface, and the yarn varies in thickness along its length. The yarn breaks at a thin, weak place, so its recorded strength is the strength of a poor section, not of the average section.

The practical consequence is a rule of specification: never promise a yarn tenacity by dividing a fibre catalogue value by a small allowance. The translation must be demonstrated on spun yarn, for the actual blend, count and twist in question.

3. The bundle and lea effect — progressive failure

Test fibres or yarns in a parallel bundle — as a lea is tested — and a second effect appears. The bundle does not fail when the average member would fail; it fails progressively. The weakest and most extensible-mismatched members break first, each break throws load onto the survivors, and the peak bundle load is reached somewhere between the strength of the weakest member alone and the simple sum of all members. Two conclusions follow.

First, a bundle result is always lower than the number of members multiplied by their mean single strength, and the gap widens as the variation among members increases. Uniformity is therefore a strength property in its own right: two yarns of equal mean single-yarn strength but different CV% give different lea and different fabric results. Second, bundle testing hides weak places, as shown in TE-3103 for the lea, because neighbours carry a broken member's load. Inside a fabric the same sharing operates — and, as the next section shows, the fabric adds help of its own.

4. Fabric assistance in the strip test

Now assemble the yarn into a woven fabric and test a ravelled strip (ISO 13934-1, Topic 3). A simple prediction multiplies the number of threads in the strip by the mean single-yarn breaking load. The measured fabric strength is regularly higher than that prediction. The excess is called **fabric assistance**, and it arises from the structure:

- **Transverse pressure at interlacings.** Where a warp thread crosses a weft thread, the two press on each other. Under load the crossing threads tighten that pressure, gripping each warp thread at every interlacing and discouraging the fibre slippage that weakens a free yarn.

- **Load sharing.** A locally weak warp thread is not alone: friction at its interlacings transfers part of its load to its neighbours and to the weft system, so the weak place is bridged instead of becoming the failure origin it would be in a single-yarn test.
- **Crimp interchange under load.** As the strip extends, warp crimp is pulled out and weft crimp increases; the weft is drawn tighter around the warp, increasing the gripping pressure just when it is most needed.
- **Restriction of the weak-place effect.** The effective free length of yarn between grips inside a fabric is the short distance between interlacings, not the full gauge length, and short lengths are on average stronger than long ones because they contain fewer weak places.

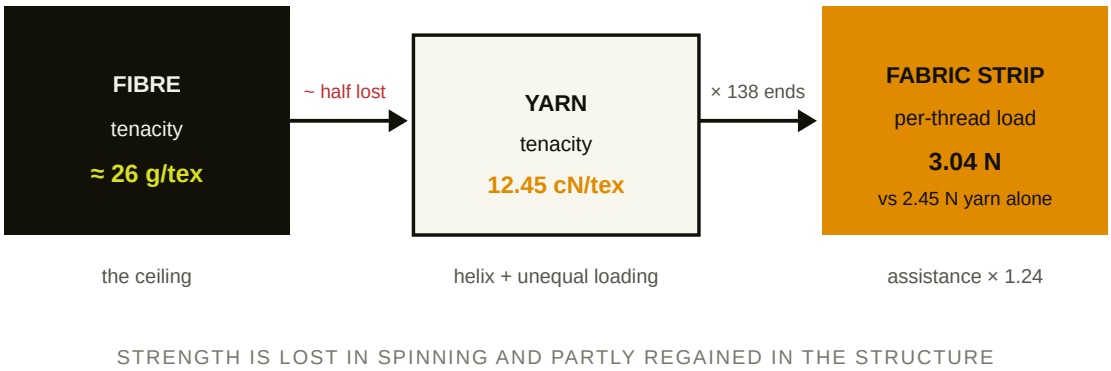


Figure 1 — The strength chain. Fibre tenacity is roughly halved in the yarn, but each thread inside the fabric strip carries more than it carried alone, because of fabric assistance.

WORKED EXAMPLE — FABRIC ASSISTANCE FACTOR

Fabric from Topics 1 and 3: warp density 70 EPI, so a 50 mm strip contains 70 × (50 / 25.4) = 70 × 1.969 ≈ **138 ends**.

Single warp yarn mean breaking load, tested alone: 245 cN = 2.45 N.

Predicted strip load = 138 × 245 cN = 33,810 cN = **338 N**.

Measured strip strength (Topic 3): **420 N**.

Fabric assistance factor = 420 / 338 = **1.24** — the fabric carries about 24% more than its threads promised alone.

Per-thread contribution in the fabric = 420 / 138 = **3.04 N**, against 2.45 N for the same yarn tested singly.

Assistance is not a fixed constant. It is larger in weaves with frequent interlacing, in fabrics whose crossing threads grip firmly, and where yarn variation gives sharing something to correct. It is smaller — and can fall below the simple prediction — when finishing damages the yarn, when the weave lets

threads slip without gripping, or when the strip breaks progressively from an edge fault. The factor is a diagnostic: a mill that knows its normal factor for a construction can spot a structural or finishing change from a single strip result.

5. Weave, thread density and finishing effects

FACTOR	EFFECT ON FABRIC STRENGTH	REASON
More threads per unit width	Strip strength rises, roughly in proportion	More load-bearing threads in the specimen
More frequent interlacing (plain vs long-float weave)	Assistance rises; threads are gripped more often — but very frequent interlacing also crimps threads more and can lower yarn condition	Pressure and load sharing at crossings
Higher crimp in the loaded system	Higher initial extension, delayed load rise; breaking load little changed if yarn is undamaged	Load first straightens the wave
Long floats	Lower assistance; threads behave more like free yarns, surface threads abrade first	Fewer gripping points per thread
Harsh bleaching / chemical damage	Strength falls at every link	Fibre polymer degraded — the ceiling itself drops
Resin / easy-care finishing	Tensile and especially tear strength commonly fall	Fibres are crosslinked and embrittled; thread mobility at a tear tip is reduced
Singeing, cropping, abrasion in finishing	Surface strength and tear resistance fall	Material is removed from thread surfaces

Two cautions complete the picture. First, density helps tensile strength but can hurt tear strength, because tightly packed threads cannot bunch at a tear tip (Topic 3) — optimising one strength can de-optimise another, and the specification must say which failure mode governs the end use. Second, a finish is judged on the finished fabric, not on the grey: a construction that passes in the grey state and fails after resin finishing was never a passing construction.

6. Implications for specification

- Specify at the level where control is possible: fibre properties for purchasing, yarn tenacity and CV% for spinning, fabric strength for the final article — and keep the three sets of limits mutually consistent through the translation factors measured on your own production.

- Use the fabric assistance factor as a routine check. A falling factor at unchanged yarn strength points to weave, density or finishing drift.
- Remember variation. Fabric strength follows yarn CV%: improving uniformity raises fabric strength without buying stronger fibre, often the cheapest strength available.
- Test the weakest direction and the governing failure mode for the end use; a chain is judged by the link that actually carries the service load.

Glossary

Translation efficiency — proportion of fibre tenacity realised as yarn tenacity. **Helix angle** — angle of a surface fibre to the yarn axis, set by twist. **Progressive failure** — sequential breaking of members of a bundle with load redistribution. **Fabric assistance** — excess of measured fabric strip strength over the sum of its threads tested singly. **Assistance factor** — measured strip strength divided by the predicted sum-of-threads strength. **Weak place** — a local section of abnormally low strength that initiates failure. **Interlacing** — a point where warp and weft change sides in the fabric.

Key points

• Yarn realises roughly half of fibre tenacity — about 26 g/tex fibre giving 12.45 cN/tex yarn in the reference cotton system — because of helix geometry, unequal loading and fibre ends.
• Bundles fail progressively; uniformity is itself a strength property.
• Fabric assistance arises from transverse pressure, load sharing and crimp interchange at interlacings.
• For the worked fabric: prediction 338 N, measurement 420 N, assistance factor 1.24, per-thread contribution 3.04 N.
• Weave, density and finishing move fabric strength independently of yarn strength — and can move tensile and tear strength in opposite directions.

Practice

1. A fabric has 64 EPI. Calculate the number of warp ends in a 50 mm ravelled strip, taking 50 mm as 1.969 inches.
2. The ends from Question 1 have a mean single-yarn breaking load of 260 cN. Calculate the predicted strip strength in newtons.
3. The measured strip strength for Question 2 is 390 N. Calculate the fabric assistance factor and the per-thread contribution in the fabric.
4. A resin finish leaves yarn strength unchanged but fabric tear strength falls sharply. Explain the fall using thread mobility, and state why the strip tensile result may fall only slightly at the same time.

Answer key

1. Ends in strip = 64 × 1.969 = 126.0, so **126 ends**.
2. Predicted load = 126 × 260 cN = 32,760 cN = **327.6 N**, about 328 N.
3. Assistance factor = 390 / 327.6 = **1.19**. Per-thread contribution = 390 / 126 = **3.10 N**, against 2.60 N for the yarn tested alone.
4. Resin crosslinks the fibres and locks the threads at their interlacings. At a tear tip the threads can no longer slide and bunch to share the concentrated load, so the tear advances thread by thread at much lower force. In a strip test the load is already shared across all threads in parallel, so loss of mobility matters less; the tensile fall is then only the direct embrittlement effect on the yarn, which in this case is small.

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